INDIA PLACE INTELLIGENCE PLATFORM

Development implementation plan · architecture · data strategy · staged production roadmap

Vision: an open-source-first, zero-planned-spend explorer where a person can search for or click any supported place in India—including remote areas—and see source-backed information about population, public health, facilities, education, policy, history, culture, weather, disasters and environmental change.

For offline planning and agent-led implementation · Version 1.0 · 18 October 2026

This is a plan, not a claim that APIs, licences, live feeds, satellite access or hosting have already been verified.

1. Executive summary

The product combines map exploration with a one-touch place profile. A user can pan/zoom without knowing a place name, search by name or coordinates, resolve the administrative hierarchy, and inspect independently sourced information panels. Health is the priority. Weather, natural disasters and satellite/environmental analysis are later phases.

Non-negotiable truth: the platform must not hallucinate. Each factual claim needs a source, geography, reference period, publication/retrieval date and quality status. Missing data is “unavailable”, not zero. Historical data is not “live”. A district-level value must not be presented as village-level measurement.

Reality checks

- **100% reliability cannot be guaranteed.** Public data can be delayed, incomplete, revised or unavailable. Show freshness, uncertainty and gaps.
- **Free/open-source software does not make every API free.** Check terms, quotas, attribution, authentication and redistribution rights for each dataset.
- **Satellite imagery is not live video.** Revisit intervals, processing delay, clouds and resolution limit what can be inferred. It cannot reliably reveal individual activity.
- **ISRO-first is a sourcing preference, not an assumed endpoint.** Verify access to each ISRO/NRSC/Bhuvan product. Do not assume a universal public API or unrestricted high-resolution real-time imagery.
- **■0 is a design target, not a hosting guarantee.** Start local-first; free tiers may sleep, enforce quotas or change policy.

2. Product experience and scope

Core interactions

1. **Explore:** pan, zoom and inspect the map at national, state, district, block/tehsil, village and point levels where supported data exists.
2. **Search:** search place names, aliases, PIN codes or coordinates; offer disambiguation for repeated names.
3. **Click:** click a boundary or point; if the name is unknown, select coordinates or draw an area and show the nearest supported administrative units.
4. **Layer controls:** boundaries, population, health indicators, facilities, schools, roads, weather, alerts and satellite/land-cover imagery as available.
5. **Evidence view:** each data card shows source, period, geographic level, last update/retrieval and limitations.
6. **Offline use:** bookmark places and export reports; cache the app shell and selected data where legally and technically possible.

Place profile panels

Panel	Content	Caution
Overview	Place name, hierarchy, coordinates/bounds, map context	Label approximate/inferred geography.
Population	Population, density, demographics, language/household indicators where published	Show census/survey year; never imply live population.
Health (priority)	Disease bulletins, available indicators, facilities, prevention guidance	Separate official alert, historical cases, modelled risk and general guidance.
Public services	Health centres, schools, water/sanitation, roads, schemes	A listing does not prove a service is operational today.
Education/policy	Schools, indicators, official policy and scheme material	A national policy is not proof of local implementation.
History/culture	Historical context, languages, heritage and local sources	Cite claims; avoid stereotypes.
Environment/satellite	Land cover, vegetation/water changes, flood/fire context, dated scenes	Show acquisition date, resolution, cloud cover and delay.
Sources/gaps	Source records, missing fields and conflicting values	Never interpret missing data as no problem.

3. Recommended technology stack

Use Python/FastAPI initially because the project is integration- and geospatial-data-heavy. Keep the backend modular; consider Rust only later if measured performance or deployment constraints justify a separate service. Avoid building two backends at the start.

Layer	Recommended choice	Reason / constraint
Frontend	TypeScript + React + Vite; installable PWA	One responsive codebase for Android, Windows and Linux browsers. PWA is not identical to native app.
Mapping	MapLibre GL JS	Open-source map renderer; smooth pan/zoom and tilt/3D-style views where style/data support it.
Map data	OpenStreetMap-derived data; verified open boundary datasets	OSM is community mapping, not an official complete gazetteer. Attribution/ODbL and tile-service terms apply.
Terrain/3D	Compatible elevation source; evaluate CesiumJS later if justified	Do not promise high-resolution global 3D before data, licence and performance are verified.
Backend	Python 3.12+, FastAPI, Pydantic, httpx	Typed API contracts and asynchronous provider integrations.
Geospatial DB	PostgreSQL + PostGIS; GeoJSON; Shapely/GeoPandas for batch work	Spatial matching, point-in-polygon and geometry validation.
Local storage	SQLite for local/small deployment; PostgreSQL for shared use	Keep local-first and portable; database is not a data source by itself.
Caching/jobs	HTTP cache/TTL; scheduled scripts first; queue only when needed	Avoid unnecessary infrastructure and paid managed services.
Frontend data	TanStack Query or equivalent; Zod validation	Loading/error states, retries, stale status and runtime schema checks.
Charts	Apache ECharts or other licence-reviewed OSS charting	Only chart time series when dated observations exist.
Testing	pytest, HTTPX test client, Vitest, Playwright	Fixtures and contract tests; test no-data, stale, conflicting, malformed and failed feeds.
Packaging/CI	Local Python tooling; optional Docker Compose; GitHub Actions free tier subject to quota	Docker must not be mandatory for every user; document Windows/Linux setup.

Architecture

TypeScript PWA → **FastAPI** → place resolver + source adapters → schema validation/normalization → provenance-aware cache/database → profile response. Background jobs refresh permitted datasets. A provider failure must not break the map or unrelated profile sections.

Suggested repository structure

`apps/web/` (UI/map/PWA); `services/api/` (FastAPI routes); `services/ingestion/` (adapters/refresh); `packages/contracts/` (schemas/types); `data/fixtures/` (small licensed test fixtures); `docs/` (source register/runbooks); `tests/` (unit, contract, integration, end-to-end).

4. Data sources and licensing strategy

Use a source registry and independent adapters. Fallback only when source, geography, time period, unit and definition are comparable. If not comparable, show separate records rather than silently substituting a different metric.

Domain	Candidates to verify	Important checks
Geography/place search	Survey of India and other official geospatial publications; Census administrative tables; OpenStreetMap/Nominatim or documented geocoders	Access, terms, rate limits, attribution, boundary vintage. Public geocoders are not unlimited.
Population	Census of India / Office of Registrar General and Census Commissioner; official surveys and open-data catalogue	Reference year, definitions, geographic level; no undisclosed interpolation.
Health/disease	MoHFW, NCDC bulletins, National Health Mission, IDSP materials where public, state health departments, official open-data catalogue	Public bulletin may not mean live API. Use permitted dated/manual ingestion if necessary.
Facilities	Official facility directories, state portals, Health Facility Registry or other documented endpoints where access allows	A listing is not proof of present operation, stock or accessibility.

Domain	Candidates to verify	Important checks
Education/schemes	Ministry of Education, UDISE+ publications, state education departments, official scheme documents	Distinguish policy existence from evidence of local implementation.
Weather	IMD public products/services where accessible; documented open weather feeds as secondary source	Verify API terms, coverage, attribution, update frequency and forecast vs observation.
Disasters	NDMA, NDRF/state disaster authorities, IMD warnings; NASA/NOAA public products where useful	Capture issuer, issue time, valid-until time and affected geography.
Indian satellite data	ISRO/NRSC products and Bhuvan where applicable	Verify each product's API/download access, licence, latency, resolution and redistribution rights.
Satellite fallback	Copernicus Sentinel services; NASA Earthdata/other suitable public science products	Check product licence, attribution, resolution, revisit, clouds and date.
History/culture	Archaeological Survey of India, official district/state publications, archives, universities and reputable local institutions	Cite individual claims; use community sources as leads, not sole authority for high-impact facts.

Source verification checklist

- Record publisher, canonical URL, documentation, licence/terms, attribution, access mode, quotas, schema, update cadence and retirement policy.
- Test a small sample before designing around a source; save a legal sample fixture and retrieval date.
- Assign status: VERIFIED, VERIFIED WITH LIMITS, MANUAL ONLY, UNAVAILABLE or NOT YET TESTED. Never call untested sources integrated.
- Prefer primary government sources for health alerts/statistics; community sources supplement rather than replace official evidence.
- Use aggregated/public-health data only. Do not scrape private patient data, bypass authentication or evade access controls.

5. Anti-hallucination and data integrity

The system is a source-grounded information tool, not an oracle. The data layer is deterministic and auditable. An optional language model may summarize validated records but must not fill gaps with invented figures, outbreaks, facility status or policy outcomes.

Canonical observation record

For each fact store fields equivalent to: `field_name, value, unit, place_id, geometry_id, source_id, source_url, publisher, source_record_id, reference_period_start/end, published_at, retrieved_at, valid_until, licence, transformation_version, quality_status, notes`. Preserve original payload or a reproducible reference where terms permit.

Required display states

- **Current/updated:** show the actual source date and valid period.
- **Historical:** label the past period; never dress it up as live.
- **Stale:** show the last known value only with a prominent stale label.
- **Unavailable:** no permitted usable record; never convert to zero or “no outbreak”.
- **Conflicting:** display source values, dates and definitions with a clear note.
- **Approximate geography:** state whether data is district-, block- or village-level.
- **Unverified community report:** label unconfirmed and moderate; never present as official alert.

Fallback procedure

1. Request primary provider with bounded timeout, retries and rate-limit handling.
2. Validate schema, date, geography and units before accepting a response.
3. Use only pre-approved compatible alternatives; do not blend incompatible definitions.
4. If none works, return a cache entry with its age/stale status or “temporarily unavailable”.
5. Log provider failures without secrets or personal data; use backoff/circuit breakers and respect Retry-After.
6. Expose source diagnostics to maintainers and users.

Health safety

- An outbreak card requires a dated statement/dataset from a competent public-health authority, with geography and valid period.
- Separate official alerts, historical cases, routine surveillance, forecasts/modelled risk and general prevention advice.
- Link precautions to current WHO, MoHFW, NCDC or relevant state-authority guidance, showing its date.
- No individual diagnosis or treatment instructions; the map is not a substitute for medical care.
- Do not invent emergency phone numbers. Suppress/aggregate tiny counts when there is re-identification risk.

LLM guardrails

- Give the model only validated records and metadata; every factual sentence must cite retrieved record IDs/URLs.
- If evidence is absent, say “not found in connected sources”. Reject citations that do not resolve.
- Keep raw data, deterministic calculations and narrative separate; narrative cannot modify canonical records.
- Use deterministic templates for high-stakes health alerts; summaries are optional and non-authoritative.
- Test stale records, contradictory sources, missing fields, adversarial prompts and fabricated citations.

6. Stage-wise implementation plan

These estimates are planning ranges for one developer working with a coding agent. Data access/licensing may take longer than implementation. Each stage has an exit gate; do not advance just because the app compiles.

Stage / estimate	Work	Exit gate
0. Feasibility, 3–5 days	Pick an Odisha pilot area; verify boundary/place lookup, census, official health sources, map terms and licences.	At least one viable place/boundary source, population dataset and official health source have sample fixtures and terms; blockers recorded.
1. Skeleton, 3–5 days	Repo layout, TS app, FastAPI, contracts, CI, lint/type checks, local setup.	Fresh checkout runs on Windows/Linux; baseline demo requires no API key.
2. Map/search, 1–2 weeks	MapLibre, pan/zoom, geocoder adapter, supported boundaries, coordinate selection, disambiguation.	Search/click resolves supported geography and shows uncertainty.
3. Profile/provenance, 1 week	Profile panels, source cards, timestamps, missing-data states, cache, export.	Every factual field has provenance and period; no fabricated defaults.
4. Population/services/education/history, 2–4 weeks	Integrate one verified dataset/domain at a time; normalize IDs and boundary versions.	Pilot values traced to sources; licence/attribution visible.
5. Health MVP, 2–4 weeks	Official bulletins/indicators, historical/current distinction, facilities, authority-linked precautions.	Stale/unavailable/conflicting tests pass; no claim of live surveillance without live feed.
6. Resilience/offline, 1–2 weeks	Cache, source diagnostics, saved places, report export, retries, PWA shell.	Outage tests pass; cache age visible; offline mode does not imply live updates.
7. Weather/disasters, v2, 2–4 weeks	Verify IMD/disaster products, map layer, validity windows, fallback and forecast charts.	Alerts expire correctly and show issuer/time; no fake live labels.
8. Satellite/environment, v2/3, 3–8+ weeks	Test ISRO/NRSC/Bhuvan access, imagery metadata, Sentinel/NASA fallback and change analysis.	One end-to-end workflow is permitted/licensed; scene dates/resolution/clouds shown.
9. Hardening/release, 1–3 weeks	Security, dependencies, privacy, performance, accessibility, backups, attribution, packaging.	Critical tests pass, no secrets committed, restore/install/docs verified.
10. Pilot/expansion, ongoing	Pilot Odisha region, compare against official records, collect feedback, expand gradually.	Coverage matrix states what exists, its date and known gaps.

7. Version roadmap

Version 1 — credible place intelligence

- Responsive TypeScript PWA for smartphone, Windows and Linux; search, click/coordinate selection and supported map tilt/3D-style presentation.
- Place profiles for population, health, facilities, education/services, history/culture and source metadata.
- Verified public datasets, timestamps, coverage, provenance, graceful provider failure, saved places and downloadable reports.
- Local-first operation; no paid key required for the baseline demo.

Version 2 — weather, disaster and satellite

- Weather observations/forecasts with clear station/model coverage and forecast-vs-observation labels.
- Official disaster alerts with issuer, issue/validity times, severity as defined by source and affected area.
- Indian satellite/remote-sensing products first where accessible; Sentinel/NASA or other suitable public products as documented fallback.
- Historical imagery comparisons and validated environmental indicators such as water extent, vegetation or land-cover change.
- Stale-feed panel; never claim real-time video or individual observation.

Version 3 — advanced analysis/institutional readiness

- Historical timelines and date comparisons only where time-series data supports them.
- Spatial aggregation and district/block comparison with uncertainty notes.
- Moderated user reports, abuse reporting and explicit unverified status.
- Institutional exports, role-based access and audit logs only if validated demand justifies them.
- Modelled risk indicators only with documented methodology, evaluation, bias analysis and expert review.

8. Mobile, Windows, Linux and offline

Target	Approach	Limitations
Android	Installable PWA, touch-friendly layout, low-bandwidth mode	Browser storage can be evicted; background tasks vary.
Windows	PWA/browser first; optional wrapper later	Desktop wrapper adds packaging and update burden.
Linux	PWA/browser; document supported browsers	GPU varies; provide reduced-motion/low-detail mode.
Offline	Cache app shell, bookmarks and exported reports; optional licensed offline map package	Countrywide high-detail maps and continually refreshed data cannot realistically be bundled for free.
Low bandwidth	Lazy-load layers, bound requests, compact summaries, manual retry	Avoid background satellite downloads over mobile data.

9. Zero-cost operating model

Target ■0 cash spend by default. Local use is the dependable baseline; public hosting is optional and must be evaluated against current terms/quotas.

- Choose open-source licences and maintain a dependency/licence inventory; attribution and compliance still apply.
- No paid API keys, credit card or paid SaaS required for the default development path.
- Publicly viewable data is not automatically licensed for bulk download, commercial use or redistribution.
- Free hosting may sleep, cap requests/storage/bandwidth or change terms; keep deployment portable and data exportable.
- Do not use public community tile servers as an unlimited commercial CDN. Select a compliant documented provider or consider self-hosting permitted data if resources allow.
- Costs to avoid: paid domains, managed databases, proprietary geocoding, premium tiles, satellite tasking, large egress, premium quotas and paid monitoring.

10. Security, privacy and responsible use

- Use HTTPS for public deployments; keep secrets out of Git and browser bundles.
- Validate inputs/upstream data; bound timeouts and response sizes; rate-limit and log safely.
- Minimize location collection; offer manual search/coordinates and explain any location permission.
- Do not expose personal health data or infer sensitive traits about named people; use aggregated geographic data.
- Moderate crowd reports against spam, harassment, doxxing, misinformation and malicious URLs.
- Run dependency/secret checks, use least privilege and document incident/rollback steps.
- Before operational agency use, obtain appropriate legal, privacy, public-health, security and governance review.

11. Agent-led execution protocol

The agent must work in small, reviewable stages, inspect the repository before editing, preserve unrelated work and never claim a check passed unless it actually ran.

Mandatory rules

1. Before every stage, inspect files/conventions and present a short plan and likely file changes.
2. Do not implement a source until access, licence, attribution, rate limits and data shape are documented.
3. Build one vertical slice first: place lookup → profile → one verified data domain → provenance.
4. Define typed contracts for Place, Source, Observation, Alert, DataStatus and API errors; version contracts.
5. Test valid, malformed, missing, stale, conflicting, timeout, rate-limit and outage cases with fixtures.
6. Mocks must be labelled DEMO/TEST. Never add fake outbreak pins or fabricated statistics to production UI.
7. No paid dependency, public deployment, prohibited scraping or production credentials without explicit approval.
8. Report changed files, exact commands, test outcomes, source verification, licensing concerns and known limitations.
9. Do not overwrite user work or reset unrelated files. Commit only when requested or allowed by the project workflow.

Definition of done

- Acceptance criteria map to automated tests or documented manual verification.
- Lint, formatting, type checks and relevant tests pass; failures are reported honestly.
- No fake success states or unhandled errors in intended failure paths.
- Source URL, licence, retrieval date, reference period and attribution are recorded.
- UI distinguishes facts, official alerts, history, estimates and unknowns.
- Documentation covers local run, configuration, source adapter maintenance and offline limits.

12. API design and core contracts

Endpoint (suggested)	Purpose	Behavior
GET /api/v1/places/search?q=	Search names/aliases/PIN/coordinates	Ranked matches, place type, parent geography, geometry availability, ambiguity.
GET /api/v1/places/{id}	Canonical place detail	Stable ID, names, hierarchy, geometry reference, boundary vintage and resolution method.
GET /api/v1/places/{id}/profile	Aggregated profile	Independent status per panel; one provider failure must not break whole profile.
GET /api/v1/places/{id}/observations?domain=health	Dated observations	Unit, period, geography, source, quality and freshness.
GET /api/v1/places/{id}/alerts	Official alerts	Issuer, published/valid times, affected geography and status.
GET /api/v1/sources	Source register/health	Terms, status, last successful fetch and limits; never expose secrets.
GET /api/v1/health	Application health	Distinguish system health from public-health alerts.

Contract rules

- Stable internal place IDs plus source-specific IDs; boundary vintage must be retained.
- Partial success per domain (for example, health=stale and education=available), not one all-or-nothing status.
- ISO 8601 timezone-aware timestamps; separate publisher time, observation period and retrieval time.
- Include metric definitions, units and denominators. Keep raw upstream data server-side where permitted.
- Document API versioning, pagination, errors, request IDs and cache headers.

13. Geospatial/data model

Entity	Minimum fields
Place	Internal ID, canonical name, aliases/language, type, parent ID, geometry/coordinates, resolution method, valid dates.
BoundaryVersion	Place ID, geometry, source/version/date, CRS, licence and processing notes.
Source	Publisher, name/URL, terms/licence, access mode, limits, verification state, last success, retirement notes.
Observation	Place, metric/value/unit, numerator/denominator, reference period, source, publication/retrieval times, quality/stale state.
Alert	Issuer, category, source record ID, issue/validity times, source severity, affected geometry, status and URL.
IngestionRun	Adapter version, times, fetched/accepted/rejected counts, schema version and error summary.
CommunityReport (future)	ID, approximate location, time, moderation state, evidence links, privacy-safe text, unverified label.

Geographic matching rules

- A point inside a known polygon may be assigned to that polygon, but show selected administrative level.
- Return the hierarchy when village, block, district and state polygons overlap.
- If no village boundary exists, do not invent one from a geocoder label; show coordinates and nearest known unit.
- District-level health data stays labelled district-level even when opened from a village page.
- Do not use point-in-polygon results across different boundary versions without recording the version.

14. Production readiness checklist

Area	Release gate
Data integrity	Every shipped dataset has source register, traceable sample, period/geography and no silent zero defaults.
Licensing	Map style, tiles, boundaries, imagery, datasets and libraries reviewed individually; attribution and redistribution rights confirmed.
Health safety	Official dated alerts; historical values not labelled current; prevention links authoritative; no individual diagnosis.
Resilience	Timeout/retry/backoff, rate-limit, cache/stale, partial-profile and provider-outage behavior tested.
Security	No secrets in repo/client bundle; validation, safe logs, dependency checks, least privilege and HTTPS.
Performance	Test on representative Android phone and modest Windows/Linux laptop; bounded layer loading.
Accessibility	Keyboard/touch support, contrast, labels, reduced motion and search alternative to map-only use.
Offline	Saved reports work offline; no implication that live alerts refresh without network.
Operations	Health endpoint, logs, source diagnostics, backup/restore test and rollback notes.
Documentation	Install/run/update, add/disable adapter, attribution, known limitations and source freshness documented.
Release	No public release until critical gates pass. Early demo must be clearly labelled as demo.

15. Risks and mitigations

Risk	Mitigation
No stable API	Use permitted official downloads/manual refresh; mark domain unavailable until validated.
Conflicting values	Show source-specific observations and definitions; no silent averaging.
Missing village data	Show finest supported geography; do not pretend district data is village-specific.
Free tier changes	Local-first, portable database, export/backup, configurable hosting.
Map quotas/terms	Configurable provider, compliant caching, documented fallback; never bypass controls.
Cloudy/delayed imagery	Show acquisition date, cloud cover, resolution; try alternate permitted scenes.
Crowd misinformation	Moderation and unverified labels; no automatic conversion to official alert.
AI false claims	Evidence-only summaries, citation validation, deterministic alert templates and tests.
Scope creep	Pilot one region and a few domains; add only after source/QA gates pass.

16. Suggested Odisha pilot

Select one district and a manageable set of blocks/villages only after checking source availability. Do not choose a place because a story suggests an outbreak. Choose a pilot where boundary files, population tables, health bulletins and a place-search path can actually be tested.

1. Map and supported pilot boundary with place lookup.
2. Canonical place profile, administrative hierarchy and coordinates.
3. At least one official population indicator with reference year.
4. At least one official health-domain record—or an explicit finding that no suitable machine-readable feed exists.
5. Facility layer only if its dataset/licence is verified.
6. Source/freshness panel and visible data gaps.
7. Automated tests for unavailable, stale, conflicting and malformed data.
8. Pilot report listing successes, gaps and next sources to verify.

17. Agreed direction vs open decisions

Agreed	Must still be verified
Open-source-first tools/maps; target ■0 spend.	Compliant map style/tile provider and sustainability.

Agreed	Must still be verified
TypeScript frontend; backend Python or Rust.	Start Python/FastAPI; revisit Rust only with evidence.
Smartphone, Windows and Linux.	PWA first; native wrapper only if justified.
Map browsing, search and clicking unnamed areas.	Pilot boundary coverage, geocoder and stable place IDs.
Population, health, services, education, history/culture.	Actual dataset/API and geographic resolution for each field.
Weather/disaster/satellite planned for later.	IMD/NDMA/ISRO/NRSC/Bhuvan and fallback access/terms/latency.
Indian satellite products preferred, scientific fallback.	Product-specific licence, resolution, revisit and redistribution rights.
Provider fallback for reliability.	Fallback only for comparable metrics and periods.
Agent codes; owner reviews stages.	Pilot district, release schedule and public deployment approval.

18. Master instruction for the coding agent

Project brief: Build an open-source-first, zero-planned-spend place-intelligence PWA for India. Use a TypeScript/React frontend with MapLibre GL JS and Python/FastAPI backend. Keep integrations modular. Start with one verified Odisha pilot and one end-to-end path: search/click place → resolve hierarchy → show a source-backed profile with explicit freshness and coverage. Health is the priority. Population, education, services, history/culture, weather/disaster and satellite/environment layers are staged additions, not assumed capabilities.

Non-negotiable: documented permitted data access; verified licences and attribution; no paid API/hosting requirement in the default path; no unapproved public deployment; no committed secrets; no fake live data; no invented boundaries or outbreak reports; no silent fallback across incompatible metrics; no claim of real-time satellite video; no individual health diagnosis. If a source is unavailable, say so and document alternatives.

Work in numbered stages with acceptance criteria. Before each stage inspect current files and report the plan. Implement only that stage, add tests, run checks and report exact commands/results. Preserve unrelated work. Label demo fixtures. Every fact must trace to a validated source record or documented deterministic calculation. Unknown is an acceptable result; hallucination is not.

19. Glossary

Term	Meaning
API adapter	Module that retrieves/normalizes one provider behind a common internal interface.
Provenance	Evidence of where a fact came from, who published it, its period and transformation.
Freshness/staleness	How recent a source is relative to its expected update cycle.
Administrative hierarchy	Nested geography such as state → district → subdistrict/block/tehsil → village where supported.
Remote sensing	Satellite/aircraft observation; not necessarily real-time or detailed enough to identify ground activity.
Fallback	Alternative source/cache used under documented compatibility and freshness rules.
PWA	Installable web app with limited offline capabilities subject to browser/device behavior.
Data gap	Field/geography lacking sufficiently reliable, permitted data.
Modelled indicator	Estimate from a documented method, distinct from direct observation.

20. Closing principle

The ambition is “Google Earth or Maps on steroids” by combining intuitive geographic exploration with a rich local knowledge profile. The credible path is not promising every fact instantly on day one. Build a system that clearly shows what is known, who says it, how old it is, which geography it represents, and what remains unknown—then expand through measured, testable stages.

Planning note: this blueprint does not mean provider access, API licences, live outbreak feeds, satellite endpoints or hosting quotas have already been verified. Those are explicit Stage 0 tasks.